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Exploring the reproductive decision support needs and preferences of women with cystic fibrosis ☆,☆☆

Mary M. Leech [a], Olivia M. Stransky [b], Mehret Birru Talabi [b,c], Sonya Borrero [b,c,d], Andrea H. Roe [e], Traci M. Kazmerski [b,f,*]

[a] Geisel School of Medicine at Dartmouth, Hanover, NH, USA
[b] Center for Women's Health Research and Innovation, Pittsburgh, PA, USA
[c] UPMC, Pittsburgh, PA, USA
[d] Center for Health Equity Research and Promotion, VA Pittsburgh Healthcare System, Pittsburgh, PA, USA
[e] Department of Obstetrics and Gynecology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA
[f] Department of Pediatrics, UPMC Children's Hospital of Pittsburgh, Pittsburgh, PA, USA

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ABSTRACT

Objectives: As women with cystic fibrosis (CF) live longer, healthier lives, they increasingly face decisions related to their reproductive health. This qualitative study explores their unique decision support needs and preferences to aid in the development of a CF-specific reproductive goals decision aid.
Study design: Women with CF age 18–44 years participated in individual, semi-structured, telephone-based interviews, and women with CF age 18 years and older participated in semi-structured focus group discussions (FGDs). Both explored experiences and attitudes surrounding parenthood, pregnancy, contraception, and preferences for reproductive health care provision. FGDs also explored the use, content, and format of a reproductive goals decision aid for women with CF. We transcribed interviews and FGDs and conducted content and thematic analyses using an inductive approach.
Results: Twenty women (age range 20–42 years) participated in interviews and 18 women (age range 26–63 years) participated in three FGDs. Major themes identified included: 1) CF complicates pregnancy and parenting decisions; 2) Women make contraceptive decisions within the context of their CF; 3) Women with CF prefer to receive reproductive health counseling from their CF team; 4) Women with CF desire defragmented, coordinated reproductive health care; and 5) A disease-specific reproductive goals decision aid would encourage relevant parenting, pregnancy, and contraceptive discussions.
Conclusion: Women with CF have unique reproductive health care needs and often face uncertainty and disjointed care when making reproductive health and contraceptive decisions.
Implications: This study underscores the central role of the CF team and illustrates opportunities to better support women with CF in their decisions surrounding sexual and reproductive health, including through a patient-centered, disease-specific, reproductive goals decision aid.
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1. Introduction

Due to advances in healthcare and medication therapy, people with cystic fibrosis (CF) are living longer, healthier lives [1]. With the release of highly effective modulator therapy (HEMT) that targets the basic defect in the CF protein, many more people with CF expect to live through and beyond their reproductive years at a healthier baseline, and women with CF are more often considering if, when, and how to start a family. A recent survey found that nearly 80% of adolescent and young adult women with CF intend to have future children, and pregnancy rates are increasing among women with CF [2,3]. However, pregnancy and early parenthood have been associated with increased CF-related clinic visits, more pulmonary exacerbations, and decreased quality of life [4]. This necessitates a greater understanding of the realities of family planning in the context of CF to help inform women's reproductive decisions.

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* Corresponding author.
E-mail address: Traci.Kazmerski@chp.edu (T.M. Kazmerski).

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Although primarily considered to be a pulmonary disease, CF significantly affects women's sexual and reproductive health (SRH) [5]. Women with CF hoping to become pregnant often experienced suboptimal fertility; however, HEMT may enhance their ability to become pregnant secondary to thinning of cervical mucus [3,6]. For those seeking to prevent pregnancy, contraception carries a few disease-specific concerns. While most forms of contraception are safe for the majority of women with CF, there are contraindications to estrogen-containing contraception in the presence of CF-related liver disease or long-term vascular access devices, and contraindications to injectable contraception in patients at risk for osteopenia or osteoporosis [7].

Women with CF have expressed concern about the effects of their reproductive choices on their health and lifespan [8]. Prior qualitative studies of family planning considerations in women with CF have shown that disease status and treatment regimen impacted decisions around pregnancy and contraception [9,10]. Women with CF also report confusion related to fertility and report major gaps in SRH care [9,11]. In contrast to women without CF, women with CF are less likely to use contraception and report rare receipt of contraceptive counseling [2,12].

We undertook this formative study to inform the development of a patient-facing decision aid to support women with CF through the complex process of recognizing and realizing their reproductive goals. We sought to explore the experiences and perspectives of women with CF related to their reproductive decision making through individual interviews. We also explored women's opinions related to MyPath, an existing reproductive goals decision aid for female veterans to clarify their reproductive goals [13], as a model for a CF-specific decision aid.

2. Methods

Women between ages 18 and 44 years old with a diagnosis of CF participated in individual interviews surrounding their reproductive health needs, perspectives, and experiences while living through their reproductive years. After building this knowledge base, we opened the conversation to women with CF age 18 years and older to participate in focus group discussions (FGDs) that captured the breadth of reproductive health experience over time, as well as their responses to the MyPath decision aid. FGD participants discussed MyPath's applicability to the CF population and the use and feasibility of such a decision aid in the CF care model. We intentionally included women past their reproductive years in the FGDs as we wanted the perspectives of women who had faced a range of reproductive decisions and who could look back on their experiences. FGDs allowed us to understand common, normative experiences of women with CF, highlight unique experiences of SRH care, and solicit feedback about an existing reproductive goals decision aid. This combined approach fostered a more personal, in-depth discussion of a woman's reproductive health experiences in the interview format and the discovery of shared experiences and specific MyPath feedback due to group interactions in the FGD setting.

We conducted all interviews using the telephone and FGDs using video conferencing software due to the strict infection control policies in the CF community [14]. We intentionally decreased the size of our FGDs to facilitate online synchronous discussions [15,16]. We offered all participants a USD50 gift card in gratitude for their time.

2.1. Interviews

We consecutively recruited individual interview participants from the adult program at the Antonio and Janet C. Palumbo CF Center at the UPMC Children's Hospital of Pittsburgh. Our study team approached eligible participants during their regularly scheduled clinic visit. Interested participants provided informed consent and scheduled an interview time with a research coordinator. MML, a female cis-gender medical student who does not have CF, and OMS, a female cis-gender Masters level research coordinator without CF who has extensive experience conducting qualitative research with women with chronic disease, conducted all of the interviews by phone between June and September 2019. The principal investigator (PI) (TMK) trained MML in qualitative interviewing, open-ended questioning, and active listening. The interviews explored past reproductive experiences, reproductive goals, experiences obtaining and using contraception, and the role of their CF team in their reproductive health care. We also asked women about the perceived usefulness of a disease-specific, web-based resource to help with parenthood or pregnancy decisions and/or contraceptive choice. Interviews lasted approximately 30–50 minutes and we audio-recorded and transcribed each verbatim. MML and OMS took notes throughout the interview and after its conclusion to allow for reflections on the interaction and early identification of themes. TMK is a female cis-gender CF physician and health services researcher with extensive experience in qualitative research with women with chronic disease and provided guidance throughout the data collection phase.

2.2. Focus group discussions

We recruited FGD participants through the CF Foundation's Community Voice group, an online listserv for people with CF and their family members to participate in research [17]. After we sent a recruitment email advertising the study, our study team contacted interested participants via email to determine availability and scheduling for the FGDs. We included all women who responded to the advertisement expressing interest in participation. We conducted FGDs via online video conferencing between April and May 2019 (Zoom Video Communications, Inc.). FGDs began with women introducing themselves and providing basic demographic information about themselves, their CF, and their reproductive health experiences. FGDs then explored the reproductive health experiences and decision support needs of women with CF, including questions about factors related to parenthood and pregnancy decisions as well as contraceptive choice, the role of the CF team in reproductive health, whether they perceived reproductive health providers as knowledgeable about CF, and how they navigated multiple providers when seeking reproductive health care. FGDs participants also viewed and provided feedback on a web-based reproductive health decision aid (MyPath) [14] The FGDs each lasted 2 hours and were conducted by the PI (TMK) and a co-investigator (AHR). AHR is a female cis-gender gynecologist with a specialty in family planning and clinical researcher. We audio- and video-recorded each FGD and transcribed them verbatim. MML and OMS attended each FGD and took notes to allow for reflections on the interaction and early identification of themes. TMK, MML, and OMS met in person after the FGDs to debrief the interactions and assess group dynamics and key themes.

2.3. Data analysis

A transcription firm contracted by the University of Pittsburgh transcribed the majority of the interviews and all FGDs; MML transcribed a subset of the interviews and audited a subset of completed transcripts. We had regular team meetings throughout the project facilitated by video conferencing to include off-site collaborators. During these meetings, we debriefed about the interview and FGD content and began to identify common themes and iden-

tify when we had reached thematic saturation. We were confident we had reached thematic saturation after 15 interviews; we conducted five additional interviews as confirmation. We also felt confident in reaching thematic saturation after our 3 FGDs, but did not conduct additional FGDs as confirmation.

We utilized Dedoose software to manage our data and facilitate coding. We analyzed the transcripts of interviews and FGDs separately using distinct codebooks. We used an inductive coding process in which codes were developed based on the topics that were discussed by participants [18,19]. Two coders (OMS and MLL) independently coded 20% of the transcripts of the interviews before developing a preliminary codebook. We revised the codebook during the coding of subsequent transcripts and the PI reviewed the final codebook for clarity and completeness. The two coders independently re-coded transcripts using the final codebook and adjudicated to consensus. They repeated this process for the FGD transcripts. Throughout the coding process for both interviews and FGDs, the research team met to discuss emerging themes based on the codes developed and utilized. Once we completed coding for both groups, the research team identified shared themes between the interviews and the FGDs. The study team noted several instances of unique additional themes uncovered in FGDs; we noted no clear disparate themes between interviews and FGDs. The University of Pittsburgh and University of Pennsylvania Institutional Review Boards approved this study.

Table 1
Characteristics of women with cystic fibrosis who participated in interviews ($n = 20$) and focus group discussions ($n = 18$).

	Frequency (%)
Interviews (n = 20)	
Age in years (Range)	20–42
Number of children*	
None	15 (75)
One	4 (20)
Two	1 (5)
Three	0 (0)
Race	
Caucasian	17 (85)
African-American	1 (5)
Asian	1 (5)
Hispanic	1 (5)
Focus Group Discussions (n = 18)	
Age in years (Range)	25–63
Number of children	
None	9 (50)
One	3 (17)
Two	5 (28)
Three	1 (5)
Race	
Caucasian	18 (100)

* In this table children are limited to those which women carried themselves. One woman in the study had a child through marriage in addition to children she carried in utero.

3. Results

3.1. Study participants

Twenty women (age range 20–42 years) participated in individual interviews. Eighteen women (age range 26–63 years) participated in one of three FGDs (seven, eight, and three participants, respectively). Among the interviewees, five had given birth to at least one child and, in the FGDs, nine women had given birth to at least one child. Given the pregnancy rates in the CF population, we believe that this population was generally representative of women with CF as related to parity. Demographics are noted in Table 1. Major themes and subthemes are shared below. Pseudonyms are utilized for all representative quotations.

3.2. CF complicates pregnancy and parenting decisions

The majority of participants named CF as a major factor in their reproductive decision-making. One woman described parenthood as a "second first priority" in a life where the top priority was her CF care. Many participants noted the importance of assessing their health before and after a possible pregnancy and the impact of their health on potential children. As one participant stated, "I looked at my health overall, if I would be able to have kids. I looked at my PFTs [pulmonary function testing], I looked at my doctors just for some assistance... I didn't want to have a kid and me die in childbirth... Those were my biggest concerns... just how healthy I would be afterwards to be able to raise a kid" (Bridget, 40 years old, three children, focus group). Participants often discussed the effect of potential maternal morbidity and mortality on their parenting decisions. Shanna shared, "In my eyes, it's not fair to [my child] to have a mom that's sick because we selfishly wanted another kid. To me, that's not responsible for her, it's not responsible of me" (35 years old, one child, interview).

The complexity CF added to parenthood for women with CF as well as their support systems emerged as a major subtheme. Many participants mentioned the challenge of balancing being a parent, having a career, and adhering to a CF care regimen. One woman said, "I couldn't be a mom, and a full-time employee, and have CF, and do them all well... For me it was easiest to give up the daily CF care, because I had to work, and I had to take care of my kids... I kinda wish somebody would have had that discussion [the impact of parenthood on adherence to CF care routines] with me, because I didn't realize my CF was catching up to me" (Melissa, 43 years old, two children, focus group). Kayla noted that partners and families of women with CF should be prepared to take on a larger role in raising a child, "[M]y husband knows he's gonna have to step up a lot more than maybe the average husband in terms of sharing responsibility with a baby" (25 years old, no children, focus group).

FGD participants often discussed the uncertainty of a person's health when living with CF. Despite recommended preconception planning, women discussed the lack of control they often felt over their CF health and outcomes and how this put strain on their parenthood and pregnancy decisions. One woman described this uncertainty, "Even though you can do the best possible to make yourself as healthy as possible [to prepare for pregnancy] ...You can still decline on the turn of a dime. Much of our health is left to chance even though we try to feel we have this control, we really don't" (Kelsey, 32 years old, no children, focus group).

3.3. Women make contraceptive decisions within the context of their CF

Many participants emphasized that they prioritized the impact of all contraceptive choices on their CF care. Women shared that they desired their contraceptive methods to easily integrate into their CF care routine, not vice versa. Rebecca indicated that an oral contraceptive would add on to her already burdensome regimen, "I just choose not to use the pill or patches or anything like that just 'cause I just don't like the side-effects that can come from them. I figure I have enough medicine with side-effects. I don't need to add to it" (34 years old, one child, interview). Christine shared, "I think my biggest thing [for contraceptive choice] is just any side effects and any interactions with my current medicines... Whatever birth control can work with my current medical regime" (43 years old, no children, focus group).

Many women felt they had been inadequately educated on how different contraceptive methods would interact with their CF or CF

medications. Several specifically cited concern about antibiotics affecting the efficacy of their contraception, such as, "I guess I kinda got lucky I didn't get pregnant before…I was told…that as many antibiotics as I'm on, I might as well swallow air because [the pill and the antibiotics are] gonna cancel each other out" (Holly, 40 years old, two children, focus group).

During the FGDs in particular, participants shared many concerns about the effects of hormones on CF health and expressed a lack of education on the topics. Some women shared that their menstrual cycle worsened CF symptoms, leading to mixed feelings about hormonal birth control. Kayla stated, "I wish that my doctors had talked to me a lot earlier about the impact of hormones on CF for females. I know around my period, I always feel worse with my CF" (25 years old, no children, focus group). Participants frequently voiced the preference for providers to advise women with CF on how female sex hormones and hormonal contraception may improve or worsen CF symptoms.

3.4. Women with CF prefer to receive reproductive health counseling from their CF team

The majority of participants stated that their CF team was their main health care provider. Many women indicated it was a personal preference to use their CF team as a primary care provider (PCP) due to the multi-system effects of CF. Kelsey stated, "I have trouble already going to other doctors that aren't my CF doctor because I feel even when I'm at another doctor, it's all really CF-related" (32 years old, no children, focus group). However, some participants indicated their non-CF doctors also preferred not having primary responsibility for healthcare decisions, such as Katrina, whose PCP "prefers that I…leave her out of everything and just go straight to CF doctors" (27 years old, no children, interview). Whether by the choice of the patients themselves or their other providers, women rely on their CF teams to educate them and help them make decisions about SRH.

While most participants had a women's health provider, the majority still preferred to receive reproductive health counseling from their CF team. A major reason for this preference was the perception of a lack of CF-specific knowledge by women's health providers. Jennifer felt that "OB/GYNs don't have enough knowledge of CF to really have any valuable input" (36 years old, one child, interview). Noel noted that her non-CF health providers "know that I know more than them" (23 years old, no children, interview). As a result, many women with CF integrated their gynecological care into their visits to the CF center. If they did have a gynecologist, they fostered interactions between the CF team and their outside provider.

3.5. Women with CF desire defragmented, coordinated reproductive health care

While many women sought reproductive health counseling from their CF team, they often viewed themselves as an intermediary in the concrete provision of their reproductive health care. The majority of participants shared the frustrating experience of managing aspects of their reproductive health care across their multiple providers. Kayla highlighted the stressful logistics of this when trying to become pregnant, "I feel very much in the middle. I'm working with an obstetrician, infertility doctors, a high-risk specialist, my CF team… It's six different doctors' offices that I'm trying to coordinate with, all around this idea of trying to get pregnant. It's very stressful and overwhelming" (25 years old, no children, focus group).

Several felt that there was no single provider who could provide care for them as both a person with CF and as a pregnant woman, with Jennifer noting, "It's weird because the CF team is worried about you, and… the obstetrician only knew the portion of taking care of the baby. I had to be the go-between to make sure that we were both getting care… It kinda felt like I was taking care of myself…" (36 years old, one child, interview). Samantha stated, "It's a ping pong match. You're just being bounced from doctor to doctor because nobody wants to claim responsibility" (36 years old, two children, focus group). This lack of a clear treatment plan or guidance in reproductive decisions left many participants feeling that they were self-managing their care. "It seems to come down that we're the middle man again trying to pull all the doctors' information into one, make an informed decision, and then find one who is gonna follow your lead," one participant shared (Whitney, 33 years old, no children, focus group).

Many patients whose doctors coordinated care across providers perceived their reproductive experiences as improved, such as Shanna who noted about her pregnancy, "I think just knowing that [my CF team was] there as a resource… I think that was a big help… Maternal-fetal medicine and the OB were the [main team], the CF [team] was the step right below them, but they definitely needed to be there" (35 years old, one child, interview). Other participants noted it was helpful to get referrals for reproductive health care from their CF team. Ella stated that when she wanted to become pregnant her CF team "referred me to a specialist who they refer their other CF patients that become pregnant or are trying to conceive too… I think it's important to have someone that is [familiar] with CF" (23 years old, no children, interview). Participants desired improved communication and collaboration between health care providers. Katrina illustrated this as follows, "If [my PCP or gynecologist] wanna start me on something, then I have to check with my [CF] clinic… I wish the communication between providers was a little bit easier" (27 years old, no children, interview).

3.6. A disease-specific reproductive goals decision aid would encourage relevant parenting, pregnancy, and contraceptive discussions

Most participants stated that they would use a web-based resource that provided information about CF and reproductive decision making. Many women emphasized how such a tool needed to be CF-specific, "I'd like to know more about the female reproductive system and CF in general. I would just like basic education on it. And if there was an app that had a whole bunch of information, I would probably read it" (Bianca, 27 years old, no children, interview).

Most participants desired in-person discussions with their CF providers related to reproductive decisions and emphasized that an online resource would be useful to "help spark discussions between patients and their doctors, stuff they may not even think about" (Holly, 40 years old, two children, focus group). After viewing the MyPath decision aid, Monica stated, "I think that creating a space for those conversations is very important in a non-judgmental manner in an open empathetic space. The way I saw this tool, I saw it as a conversation starter about contraceptive and family planning so that the care team and the CF adults can be on the same page. (39 years old, one child, focus group).

Women noted the importance of any CF reproductive decision aid being evidence-based and transparent; for example, "One of the things I'd like to be able to do would be click on something and get access to the medical mumbo-jumbo. I hate it when websites think they're gonna dumb things down and make it simple because I can't understand the medical jargon. I think there's a lot [of people with CF] out there that read PubMed articles and hack their way through the journal articles and stuff just to understand the latest in research and how things are working" (Carol, 63 years old, two children, focus group).

4. Discussion

This study explored the experiences, needs, and preferences of women with CF related to their family planning decisions. CF complicates women's reproductive health decisions to become a parent, become pregnant, or choose a contraception method in a variety of ways. Women often view their CF physicians as their PCPs and defer to them for comprehensive health advice, including counseling related to their reproductive health decisions. Participants frequently described the experience of fragmented reproductive health care and the desire for improved collaboration between their CF team and women's health providers. Additionally, women emphasized that a disease-specific reproductive health decision aid should encourage discussions and shared decision-making between patients and their CF team.

Due to the complex nature of the disease, people with CF receive care at specialized, accredited CF centers by teams of pulmonologists, dietitians, nurses, respiratory therapists, program coordinators, and social workers, who often work in conjunction with physical therapists, psychologists, research coordinators, and pharmacists [20]. This team plays a central role in the comprehensive healthcare of people with CF and prior work has demonstrated that many people with CF view their CF subspecialists as de facto PCPs [21]. However, while women with CF prefer talking to their CF care team about reproductive health decisions, two-thirds of CF providers are uncomfortable asking questions related to family planning, and more than half are uncomfortable providing preconception or contraceptive counseling [22]. This highlights the need for the development of patient-facing interventions, such as online educational resources and disease-specific decision aids that can educate, support, and guide women with CF as they face reproductive choices.

This formative research study identified that women with CF would value a disease-specific web-based reproductive goals decision aid. It specifically demonstrated that women with CF view such a tool as a "conversation-starter" between patients and their CF providers rather than a stand-alone tool. Use of language and content that is meaningful, specific, and relevant to the user is vital in patient-facing interventions, including decision aids [23]. In this study, women identified the importance of including scientific references and source data in any tool to be developed.

In response to this work, a group of key stakeholders, including women with CF, maternal-fetal medicine specialists, family planning providers, and CF team members, are using these data to improve patient-provider communication and inform the design of a patient-facing reproductive health decision aid for women with CF. Following the creation of the aid, pilot testing will be conducted with attention to preliminary effectiveness on the frequency and quality of reproductive health discussions between patients and providers.

Participants in this study were concerned that their reproductive health providers did not sufficiently understand CF to provide useful care. Additionally, they felt there was a major disconnect in the communication within the healthcare system (specifically between their CF and OB/GYN teams) related to family planning care. In an effort to facilitate such collaboration, a recent study from France piloted incorporating gynecologic services directly into the CF center, which improved overall rates of gynecologic care provision [24]. Such integration may raise patients' trust of SRH providers as they are affiliated with their CF care team and improve access to the full spectrum of necessary gynecologic and obstetric care. Investments in the training of women's health providers in modern CF care, including the impact of HEMT on family planning decisions, are also warranted.

Women with other chronic conditions, such as rheumatic disease and epilepsy, endorse similar concerns related to disease-specific SRH concerns [25–27]. Additionally, women with such conditions also report confusion and suboptimal knowledge related to SRH and contraception [28–30]. Patient-facing educational resources have been developed for women with many chronic conditions [31–34], but more concrete decision aids related to reproductive goals have not yet been undertaken for these populations. This work, therefore, can serve as a model for future decision aids that can be specifically tailored to help women with other chronic diseases navigate the complex SRH landscape.

This study has several limitations. Although we are confident that the themes that we identified in our FGDs and interviews reflect broader social norms and themes and have meaning beyond the study population, the qualitative nature of this study means that it is not generalizable. Regarding the elements of homogeneity among FGD participants, all were participants in the CFF Community Voice, suggesting that they are active in CF advocacy and research at national level. We did not intentionally seek homogeneity related to race for our FGDs, but, given the patient population of CF in the United States (U.S.), all were white. All participants in both interviews and FGDs are cared for at CFF-accredited centers (similar to >90% of those with CF in the U.S.). We intentionally included heterogeneity in FGD participants related to their reproductive goals and stage of life to encourage discussion across the group and comparison of different viewpoints related to SRH decisions. We also intentionally included geographically diverse FGD participants to account for possible SRH care provision differences by CF care teams and region. We recruited interview participants from one CF center and, thus, they do not reflect a geographic diversity. The study team who lead our FGDs consisted of a CF health services researcher and a family planning provider. A Masters' level research coordinator and a medical student conducted our individual interviews. As is true of qualitative research, the positionality of our study team, including gender, educational level, parity, and marital status, influenced the FGD and interview interactions. Through debriefings and regular team meetings, we attempted to understand these influences and enhance the credibility of this study.

Despite these limitations, qualitative methods provided the ability to explore patient experiences and preferences with more depth than could be obtained via quantitative methods. Participants shared personal stories to illustrate their experiences and defend their opinions, and we obtained specific, detailed feedback on an existing decision aid to inform the development of a patient-centered decision aid.

In conclusion, women with CF have unique decision support needs and preferences related to pregnancy, contraception, and parenting decisions. Women with CF face uncertainty related to the impact of family planning decisions on their overall health, and there is a lack of evidence surrounding the long-term effects of pregnancy, parenthood, and contraceptive choice in this population. Inclusion of analysis of the impact of HEMT on the interplay between CF and reproductive health is paramount. Improved collaborative care of women with CF by CF teams and women's health providers can better support this population. A disease-specific reproductive health decision aid that provides evidence-based, CF-specific information may encourage patient-provider conversations related to family planning, improving the access to care and quality of life for women with CF.

References

[1] Cystic Fibrosis Foundation. Patient Registry Annual Data Report; 2019.
[2] Kazmerski TM, Sawicki GS, Miller E, Jones KA, Abebe KZ, Tuchman LK, Ladores S, Rubenstein RC, Sagel SD, Weiner DJ, Pilewski JM, Orenstein DM, Borrero S. Sexual and reproductive health behaviors and experiences reported by young women with cystic fibrosis. J Cyst Fibros 2018;17(1):57–63.

[3] Heltshe SL, Godfrey EM, Josephy T, Aitken ML, Taylor-Cousar JL. Pregnancy among cystic fibrosis women in the era of CFTR modulators. J Cyst Fibros 2017;16(6):687–94.
[4] Schechter MS, Quittner AL, Konstan MW, Millar SJ, Pasta DJ, McMullen A. Long-term effects of pregnancy and motherhood on disease outcomes of women with cystic fibrosis. Annals ATS 2013;10(3):213–9.
[5] Frayman KB, Sawyer SM. Sexual and reproductive health in cystic fibrosis: a life-course perspective. Lancet Respir Med. 2015;3(1):70–86. https://doi.org/10.1016/S2213-2600(14)70231-0.
[6] Sergeev V, Chou FY, Lam GY, Hamilton CM, Wilcox PG, Quon BS. The extra-pulmonary effects of CFTR modulators in cystic fibrosis. Ann Am Thorac Soc. 2020;17(2):147–54.
[7] Whiteman MK, Oduyebo T, Zapata LB, Walker S, Curtis KM. Contraceptive safety among women with cystic fibrosis: a systematic review. Contraception 2016;94(6):621–9.
[8] Hailey CE, Tan JW, Dellon EP, Park EM. Pursuing parenthood with cystic fibrosis: reproductive health and parenting concerns in individuals with cystic fibrosis. Pediatr Pulmonol 2019;54(8):1225–33. https://doi.org/10.1002/ppul.24344.
[9] Traxler SA, Chavez V, Hadjiliadis D, Shea JA, Mollen C, Schreiber CA. Fertility considerations and attitudes related to family planning in women with cystic fibrosis. Contraception 2019;100(3):228–33.
[10] Kazmerski TM, Gmelin T, Slocum B, Borrero S, Miller E. Attitudes and decision making related to pregnancy among young women with cystic fibrosis. Matern Child Health J 2017;21(4):818–24.
[11] Kazmerski TM, Sawicki GS, Miller E, Jones KA, Abebe KZ, Tuchman LK, Ladores S, Rubenstein RC, Sagel SD, Weiner DJ, Pilewski JM, Orenstein DM, Borrero S. Sexual and reproductive health care utilization and preferences reported by young women with cystic fibrosis. J Cyst Fibros 2018;17(1):64–70.
[12] Roe AH, Traxler SA, Hadjiliadis D, Sammel MD, Schreiber CA. Contraceptive choices and preferences in a cohort of women with cystic fibrosis. Respir Med 2016;121:1–3.
[13] MyPath Intervention [Internet]. U.S. Department of Veterans Affairs; c2019 [cited August 2019] Available from: http://www.mypathtool.org. Accessed September 29, 2020.
[14] Saiman L, Siegel JD, LiPuma JJ, Brown RF, Bryson EA, Chambers MJ, et al. Cystic Fibrosis Foundation. Infection prevention and control clinical care guidelines. Infect Control Hosp Epidemiol. 2014;35(1):1–67. https://doi.org/10.1086/676882.
[15] Stewart K, Williams M. Researching online populations: the use of online focus groups for social research. Qualitative Res. 2005;5(4):395–416.
[16] Fox FE, Morris M, Rumsey N. Doing synchronous online focus groups with young people: methodological reflections. Qual Health Res 2007;17(4):539–47.
[17] Community Voice [Internet]. Cystic Fibrosis Foundation. Available from: https://www.cff.org/Get-Involved/Connect/Community-Voice/. Accessed September 29, 2020.
[18] Saldana J. The coding manual for qualitative researchers. 2nd ed. London: Sage; 2016.
[19] Crabtree BF, Miller WL, editors. Doing qualitative research. London: Sage; 1999.
[20] Your Care Team [Internet]. Cystic Fibrosis Foundation. Available from: https://www.cff.org/Care/Your-CF-Care-Team/. Accessed September 29, 2020.
[21] Britto MT, Garrett JM, Dugliss MA, Johnson CA, Majure JM, Leigh MW. Preventive services received by adolescents with cystic fibrosis and sickle cell disease. Arch Pediatr Adolesc Med. 1999;153(1):27–32.
[22] Kazmerski TM, Nelson EB, Newman LR, Haviland MJ, Luff D, Leichtner AM, Hayes MM, Miller E, Emans SJ, Sawicki GS. Interprofessional provider educational needs and preferences regarding the provision of sexual and reproductive health care in cystic fibrosis. J Cyst Fibros 2019;18(5):671–6.
[23] Joseph-Williams N, Newcombe R, Politi M, Durand M-A, Sivell S, Stacey D, O'Connor A, Volk RJ, Edwards A, Bennett C, Pignone M, Thomson R, Elwyn G. Toward minimum standards for certifying patient decision aids: a modified Delphi consensus process. Med Decis Making 2014;34(6):699–710.
[24] Rousset-Jablonski C, Reynaud Q, Perceval M, Nove-Josserand R, Durupt S, Ray-Coquard I, Golfier F, Durieu I. Improvement in contraceptive coverage and gynecological care of adult women with cystic fibrosis following the implementation of an on-site gynecological consultation. Contraception 2020;101(3):183–8.
[25] Wolgemuth T, Stransky OM, Chodoff A, Kazmerski TM, Clowse ME, Birru TM. Exploring the preferences of women regarding sexual and reproductive health care in the rheumatology context: a qualitative study. Arthritis Care Res 2020. https://doi.org/10.1002/acr.24249.
[26] Birru Talabi M, Eudy AM, Jayasundara M, Haroun T, Nowell WB, Curtis JR, Crow-Hercher R, White CW, Ginsberg S, Clowse MEB. Pregnancy, periods, and "the pill": exploring the reproductive experiences of women with inflammatory arthritis. ACR Open Rheuma 2019;1(2):125–32.
[27] Manski R, Dennis A. A mixed-methods exploration of the contraceptive experiences of female teens with epilepsy. Seizure 2014;23(8):629–35.
[28] Birru Talabi M, Clowse MEB, Blalock SJ, Switzer G, Yu L, Chodoff A, Borrero S. Development of ReproKnow, a reproductive knowledge assessment for women with rheumatic diseases. BMC Rheumatol 2019;3(1).
[29] Huang VW, Chang H-J, Kroeker KI, Goodman KJ, Hegadoren KM, Dieleman LA, Fedorak RN. Does the level of reproductive knowledge specific to inflammatory bowel disease predict childlessness among women with inflammatory bowel disease? Canad J Gastroenterol Hepatol 2015;29(2):95–103.
[30] Korzeniewska A, Grzelewski T, Jerzyńska J, Majak P, Sołoniewicz A, Stelmach W, Stelmach I. Sexual and reproductive health knowledge in cystic fibrosis female patients and their parents. J Sexual Med 2009;6(3):770–6.
[31] Kazmerski TM, Miller E, Sawicki GS, Thomas P, Prushinskaya O, Nelson E, Hill K, Miller A, Jean Emans S. Developing sexual and reproductive health educational resources for young women with cystic fibrosis: a structured approach to stakeholder engagement. Patient 2019;12(2):267–76.
[32] Girls with Nerve [Internet]. Ibis Reproductive Health; c2015. Available from: http://girlswithnerve.com. Accessed September 29, 2020.
[33] Center for Young Women's Health [Internet]. Available from: https://youngwomenshealth.org. Accessed September 29, 2020.
[34] Charron-Prochownik D, Ferons-Hannan M, Sereika S, Becker D. Randomized efficacy trial of early preconception counseling for diabetic teens (READY-Girls). Diabetes Care 2008;31(7):1327–30.